Vorlesung
Informationsvisualisierung

Kapitel 4
VISUALISIERUNG VON HIERARCHIEN

Lecture
Information Visualization

Chapter 4
VISUALIZATION OF HIERARCHIES

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Universität Trier – Sommersemester 2026

Next Tutorial

- Tuesday, 12.5.2026 at 10:15 in HZ 203

Chapter 4.1

Hierarchical Data

Examples of trees / hierarchical structures

Examples

General

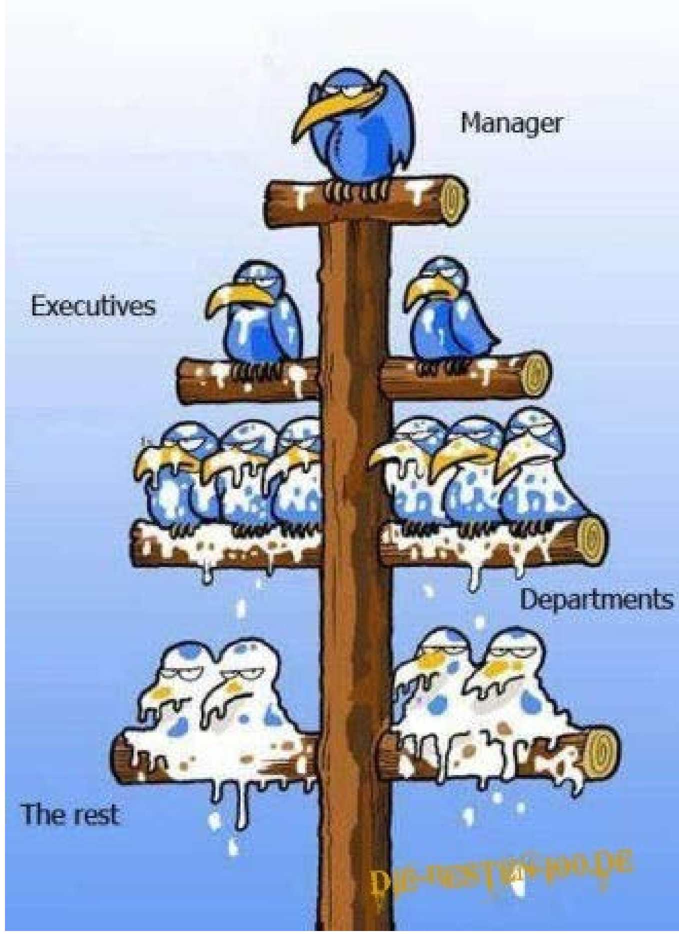
- Company hierarchy
- Family tree
- Hierarchy of categories

Natural Scienes / Geography

- River system
- Geographical region (continents - countries - regions)
- Phylogenetic trees

Computer Science

- File system
- Data structures (e.g. search trees)
- Composite Pattern
- (Single) Inheritance (OO)



Mathematical Definitions of Hierarchical Structures

- Hierarchy = Tree
- Definition: A tree $T = (V, E)$ *is a connected,* ***undirected*** *graph without cycles comprising*
  - V: a set of vertices
  - $E \subseteq V \times V$: a set of edges
- Types of vertices:
  - inner vertex: degree > 1
  - leaf: degree = 1
- In a **rooted tree** one vertex has been designated the root. (Alternative definition: **directed tree**)
- Note: In a tree there is exactly one path from the root to each vertex.

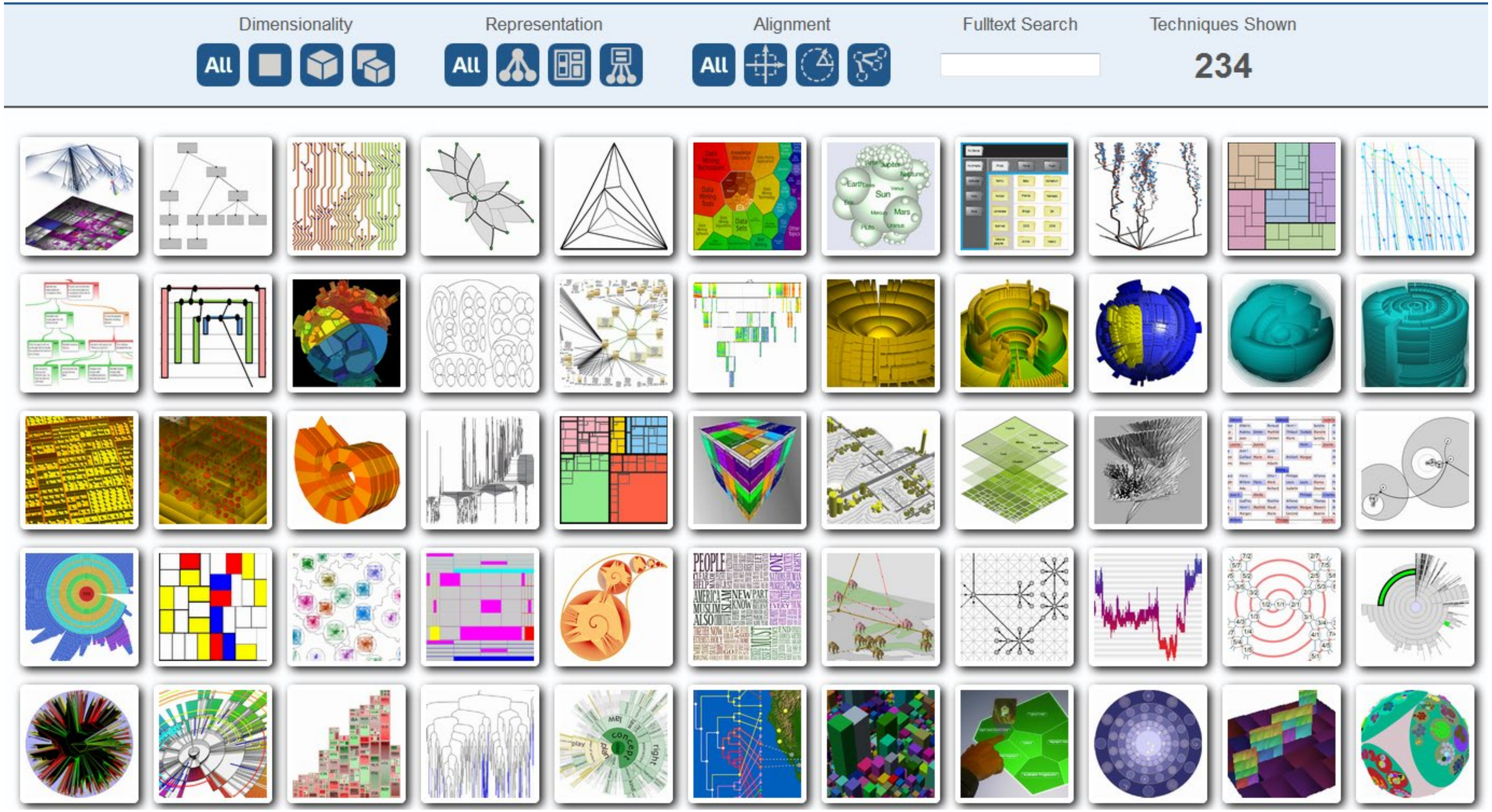
Equivalent Definitions

An undirected graph $G=(V,E)$ with $|V|=n$ vertices and $|E|=m$ edges is a tree if it satisfies any of the conditions below:

- G is connected and has no cycle. (Definition of previous slide)
- There is exactly one path between each pair of vertices.
- G is connected and $m=n-1$. (There is one edge less than vertices).
- G has no cylcle and $m=n-1$.
- G is minimal connected, i.e. G is connected, but if we removes any one edge, it is no longer connected.
- G is maximal acyclic, i.e. G has no cycle, but if we add another edge between two arbitrary vertices, there will be a cycle.

Visualization of Hierarchies

treevis.net



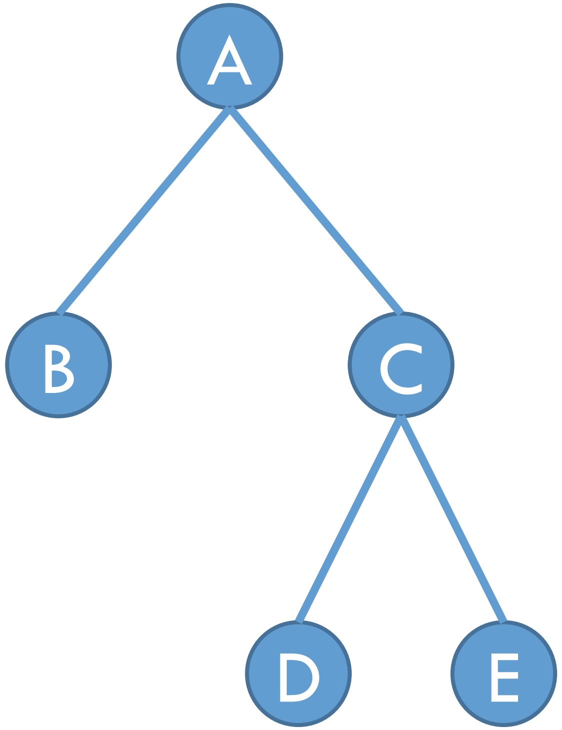
Chapter 4.2

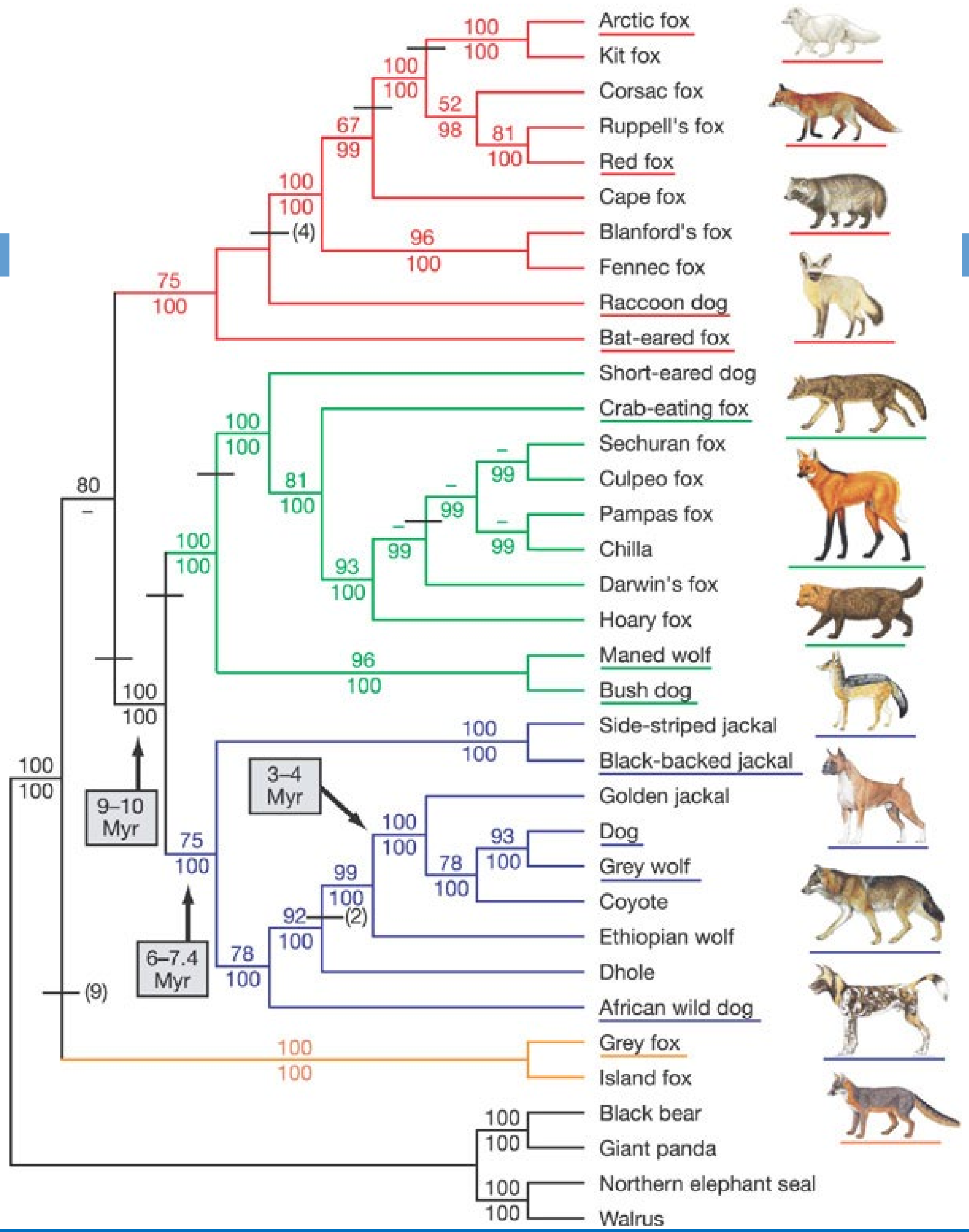
Node-Link Diagrams

Mathematical Concept → Visual Representation

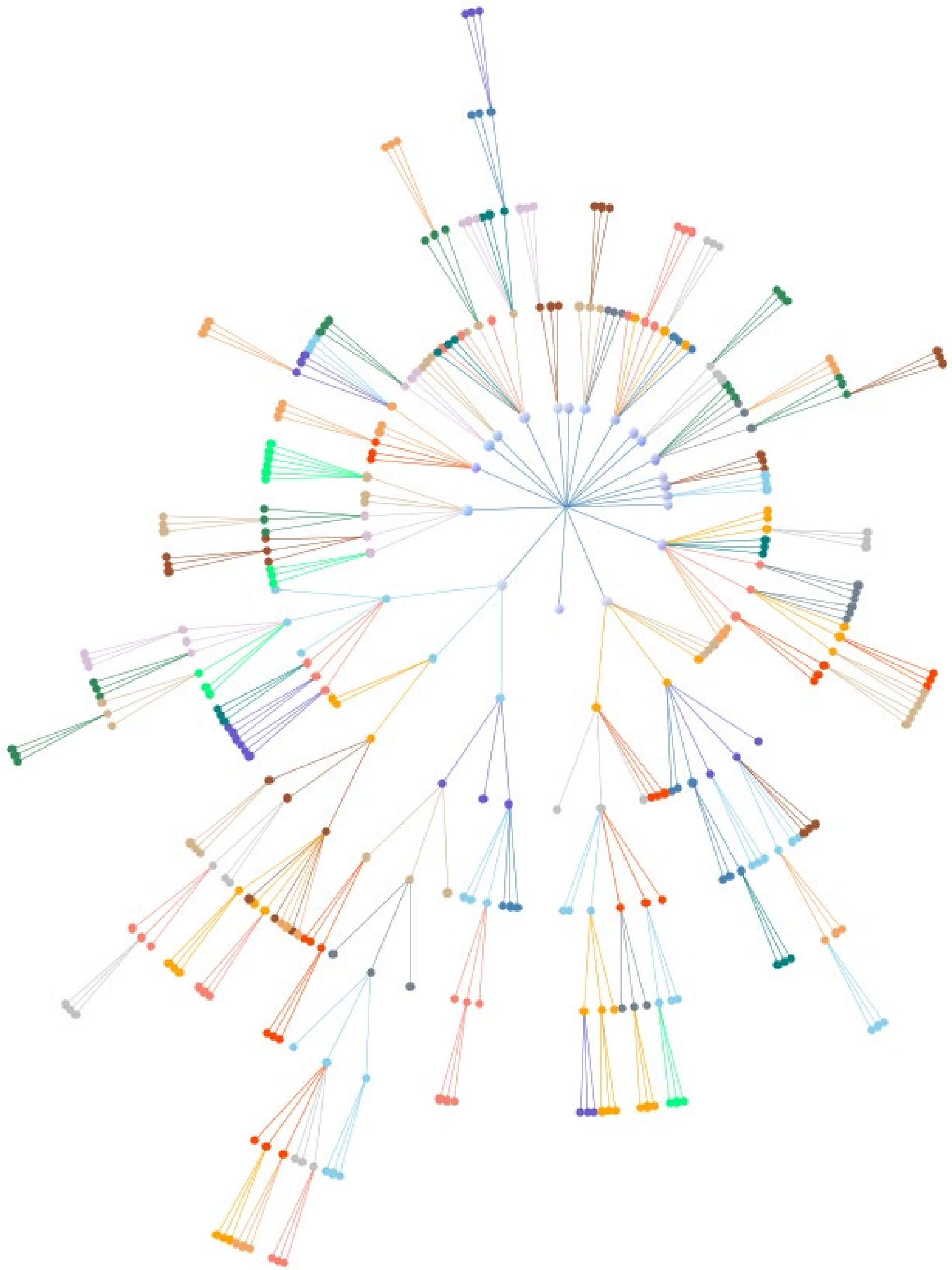
Vertex → Node

Edge → Link

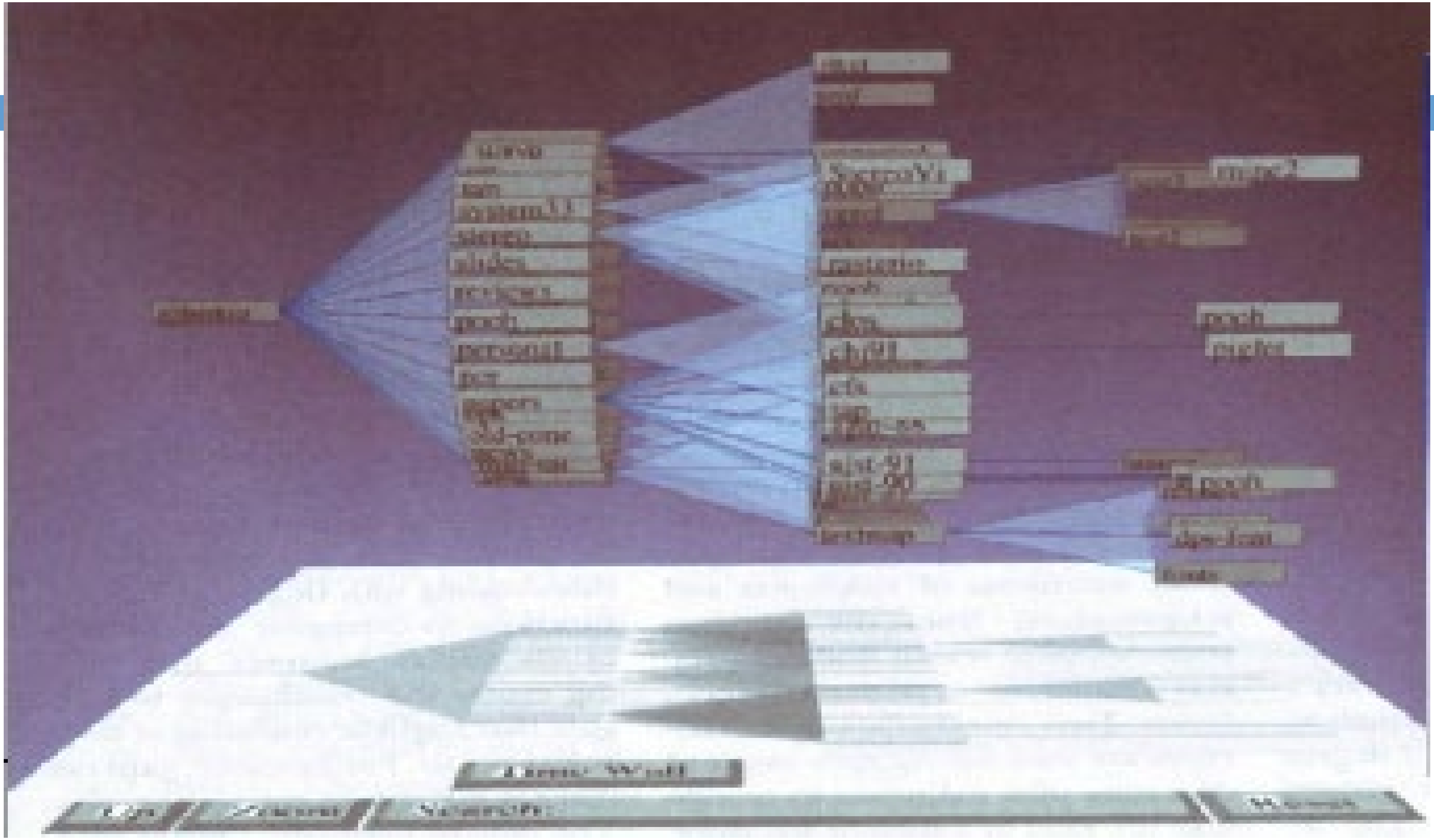




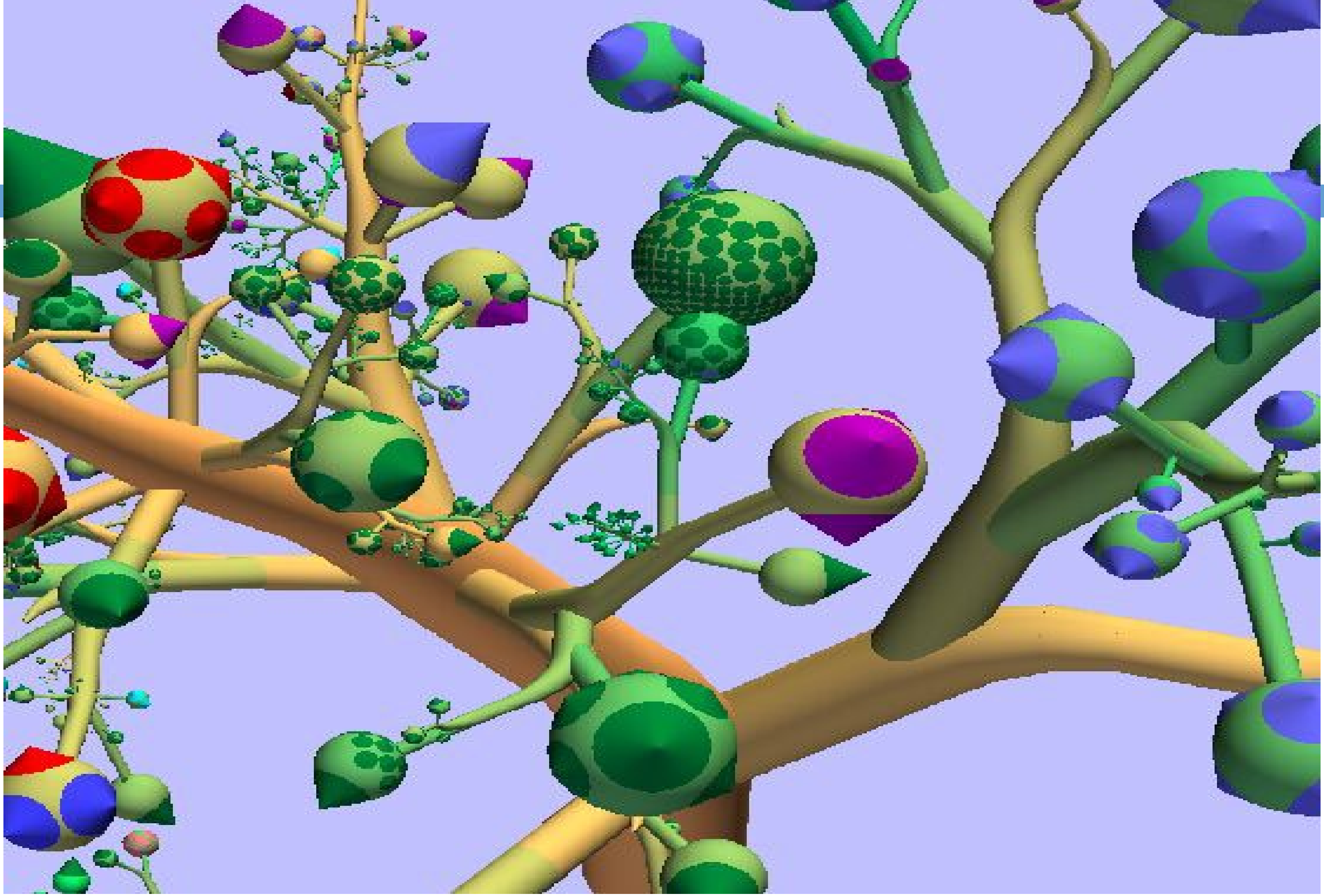
Phylogenetic Tree
(Lindblad-Toh et al., Nature, 2005)



Radial Tree
Orbidfold G2



Cone Trees; Robertson et al., 1991



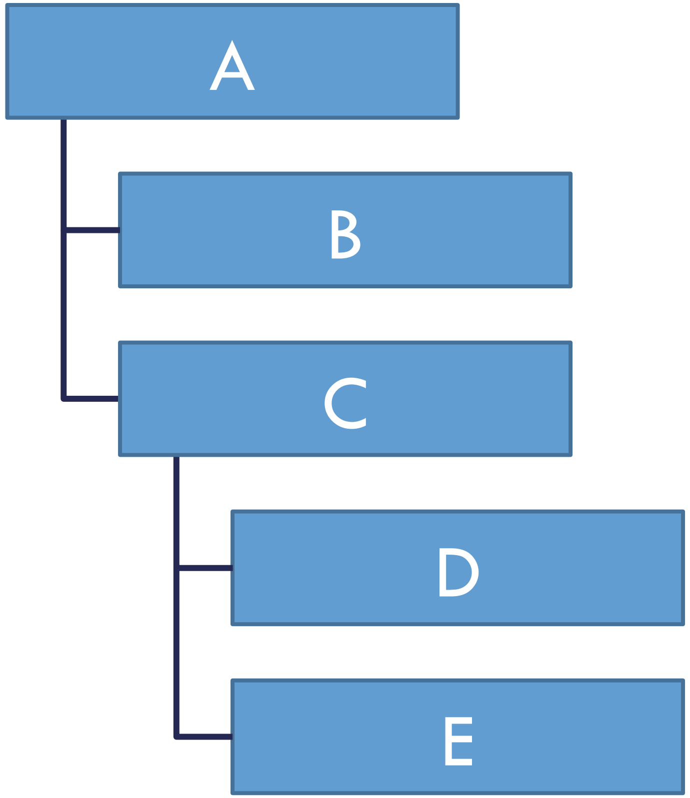
Botanical Visualization of Huge Hierarchies; Kleiberg et al., 2001

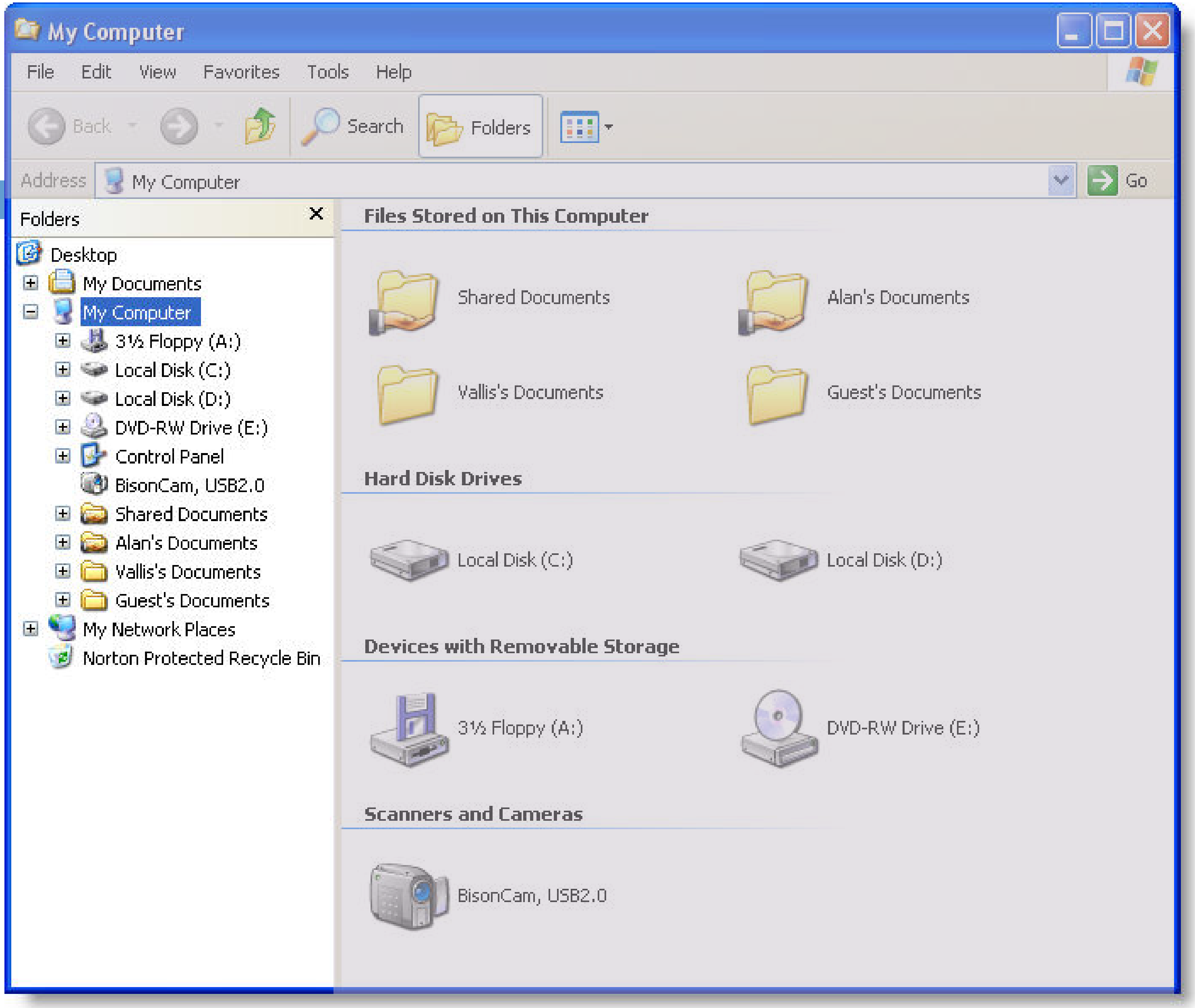
Discussion: Node-Link Representation

Advantages	Disadvantages
Intuitive	Edges require space
Hierarchy immediately recognisable	**Difficult to add labels**
Very flexible layout	Degenerated trees are difficult to represent

Chapter 4.3

Indented Outline Plot





Windows Explorer; Microsoft Windows

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```


XML File; Mozilla Firefox

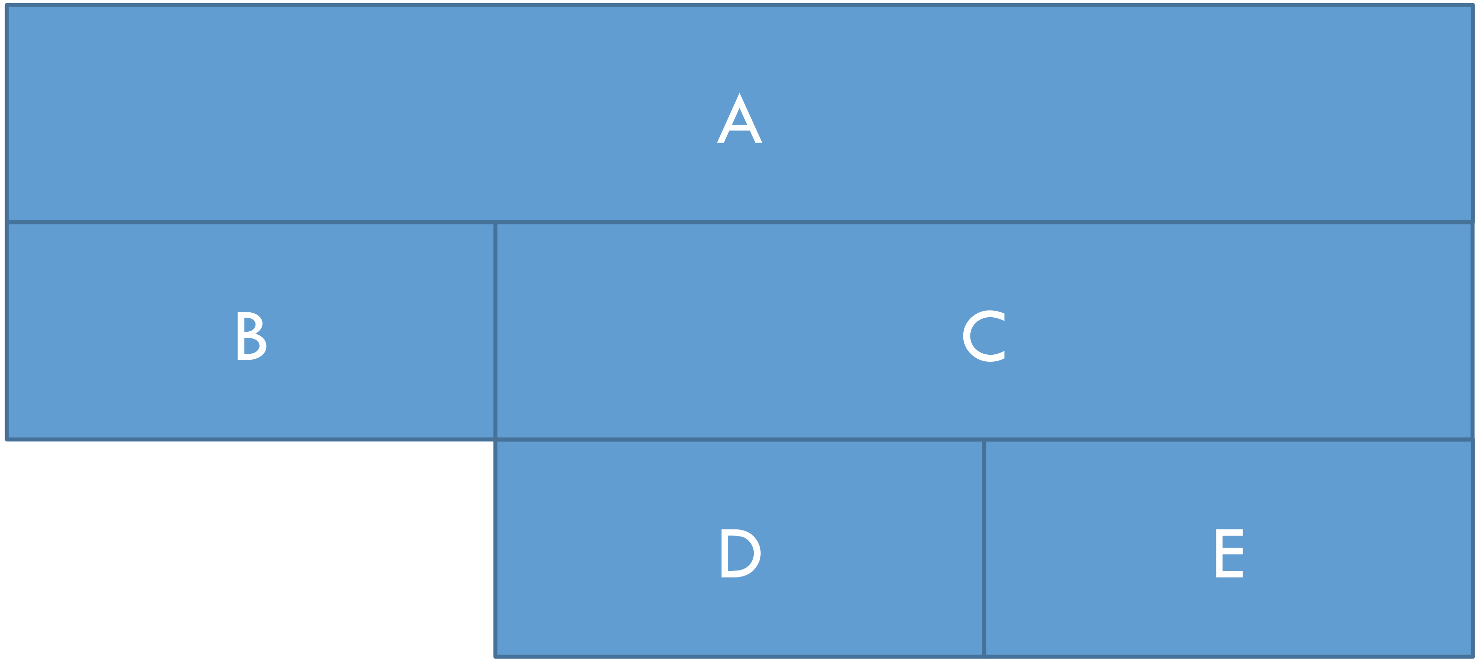
Discussion

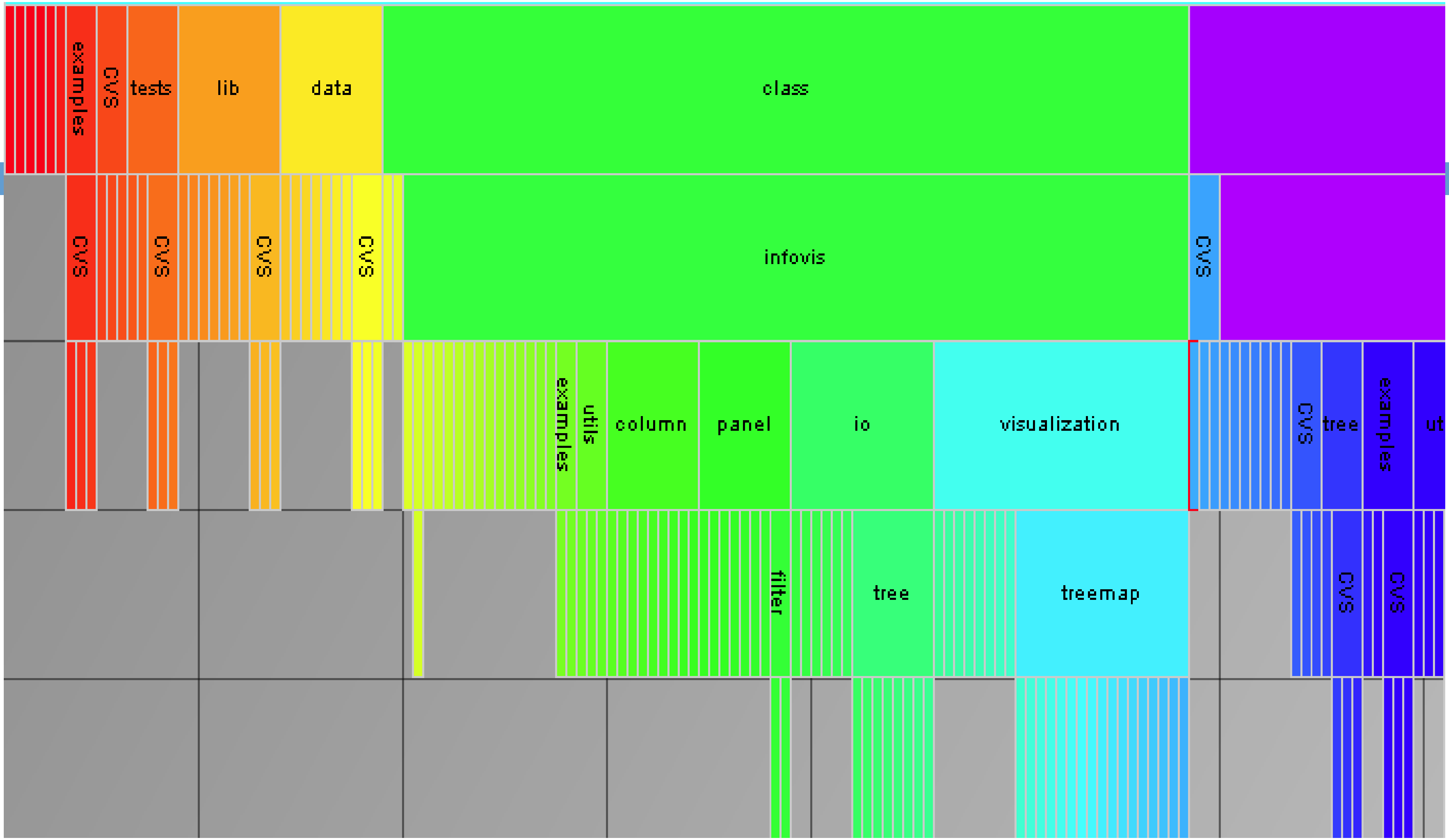
Advantages	Disadvantages
Very readable	Inner nodes require space
Easy to add labels	Somewhat inflexible layout
Familiar; used daily by many people (file explorer)	
Degenerated trees can be represented	
Hierarchy is well recognisable	

Chapter 4.4

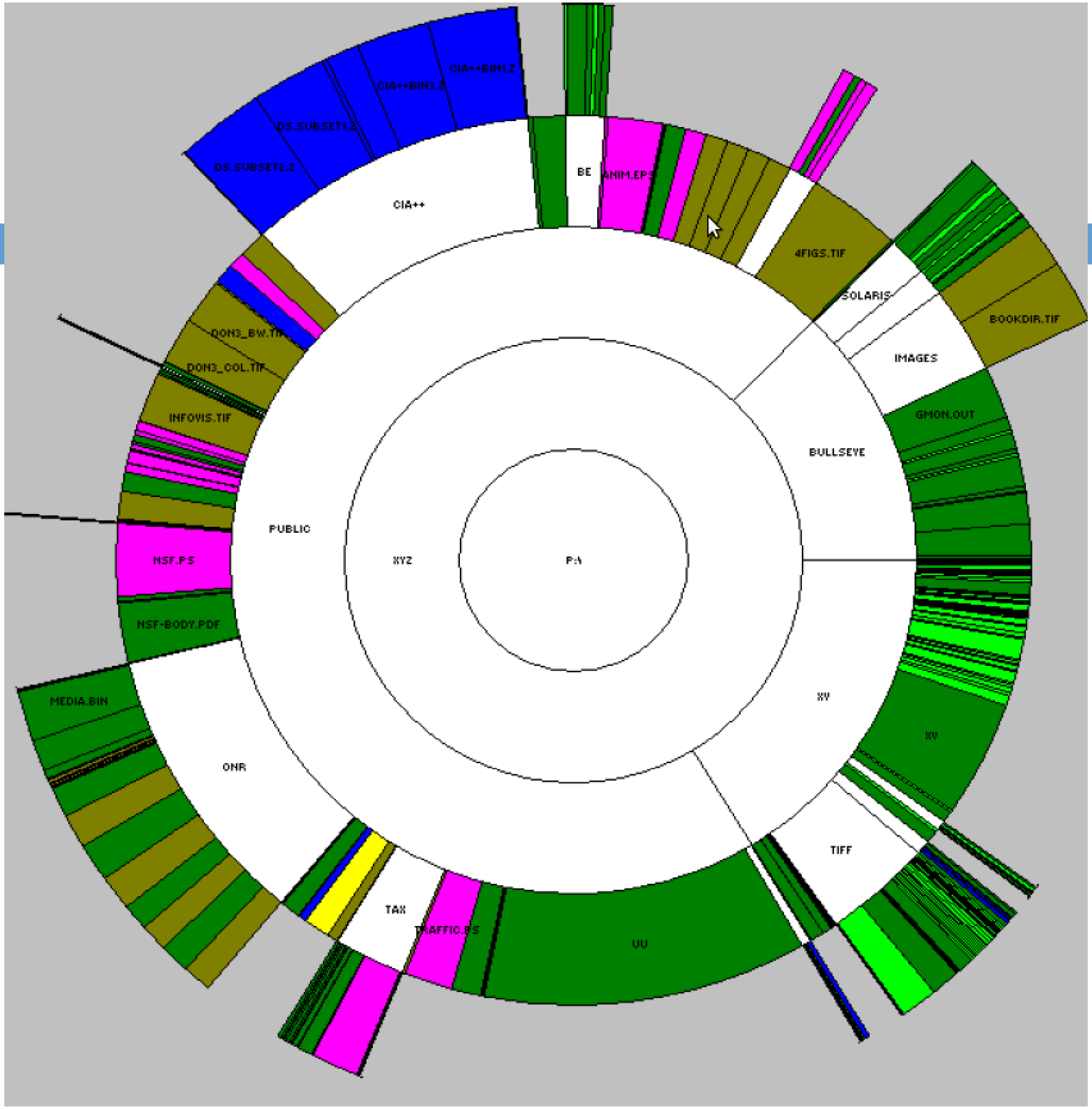
Icicle Plot

Icicle Plot

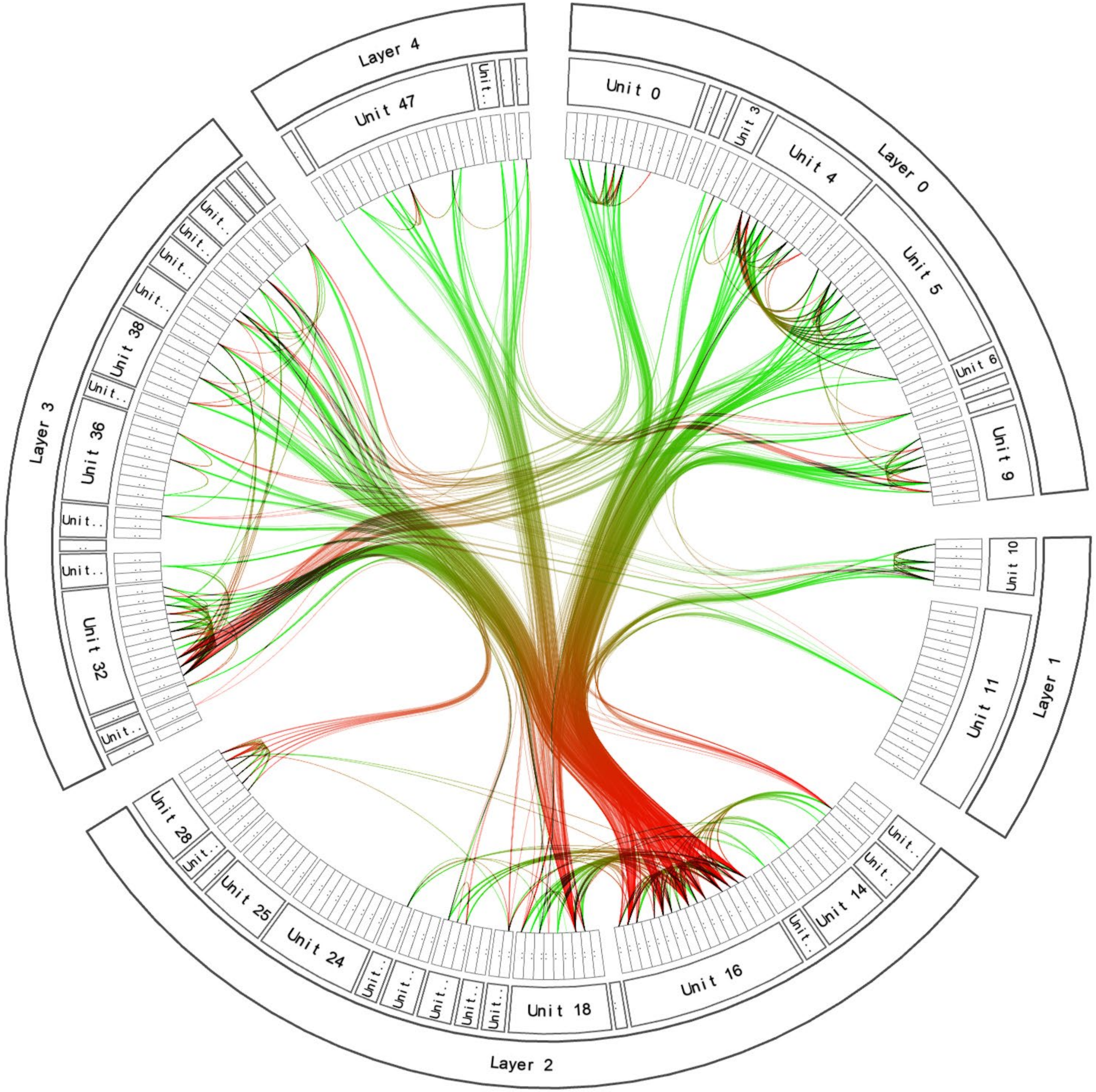




InfoVis Toolkit; Fekete, 2004



Sunburst; Stasko and Zhang, 2000



Hierarchical Edge Bundles; Holten, 2006

Discussion: Icicle Plot

Advantages	Disadvantages

Advantages

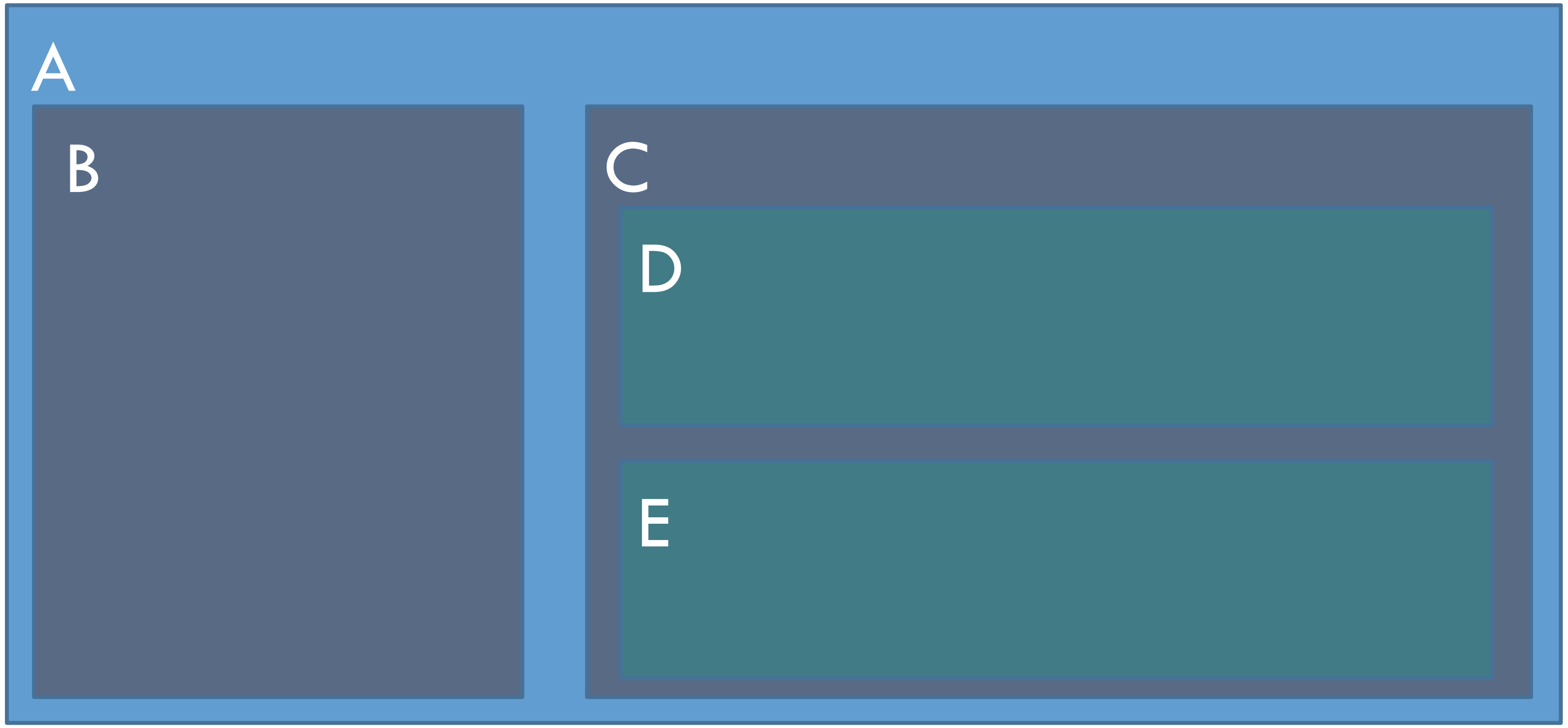
- **Easy to add labels**
- Hierarchy is well recognisable
- Flexible layout
- Uses screen space efficiently

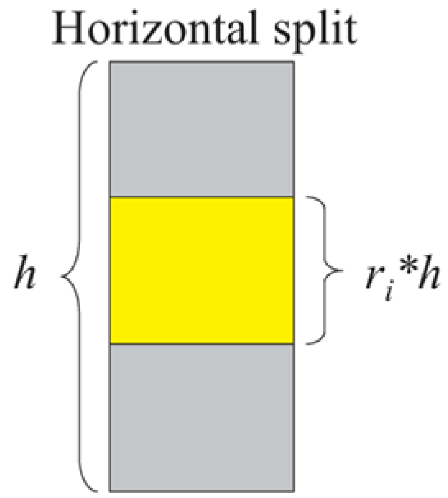
Disadvantages

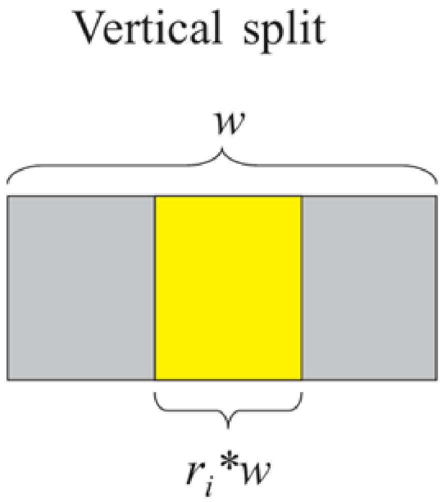
- Somewhat less intuitive
- Available width for children restricted by the width of their parents.

Chapter 4.5 Treemap

http://www.cs.umd.edu/hcil/treemap/







Recursive algorithm with two phases:

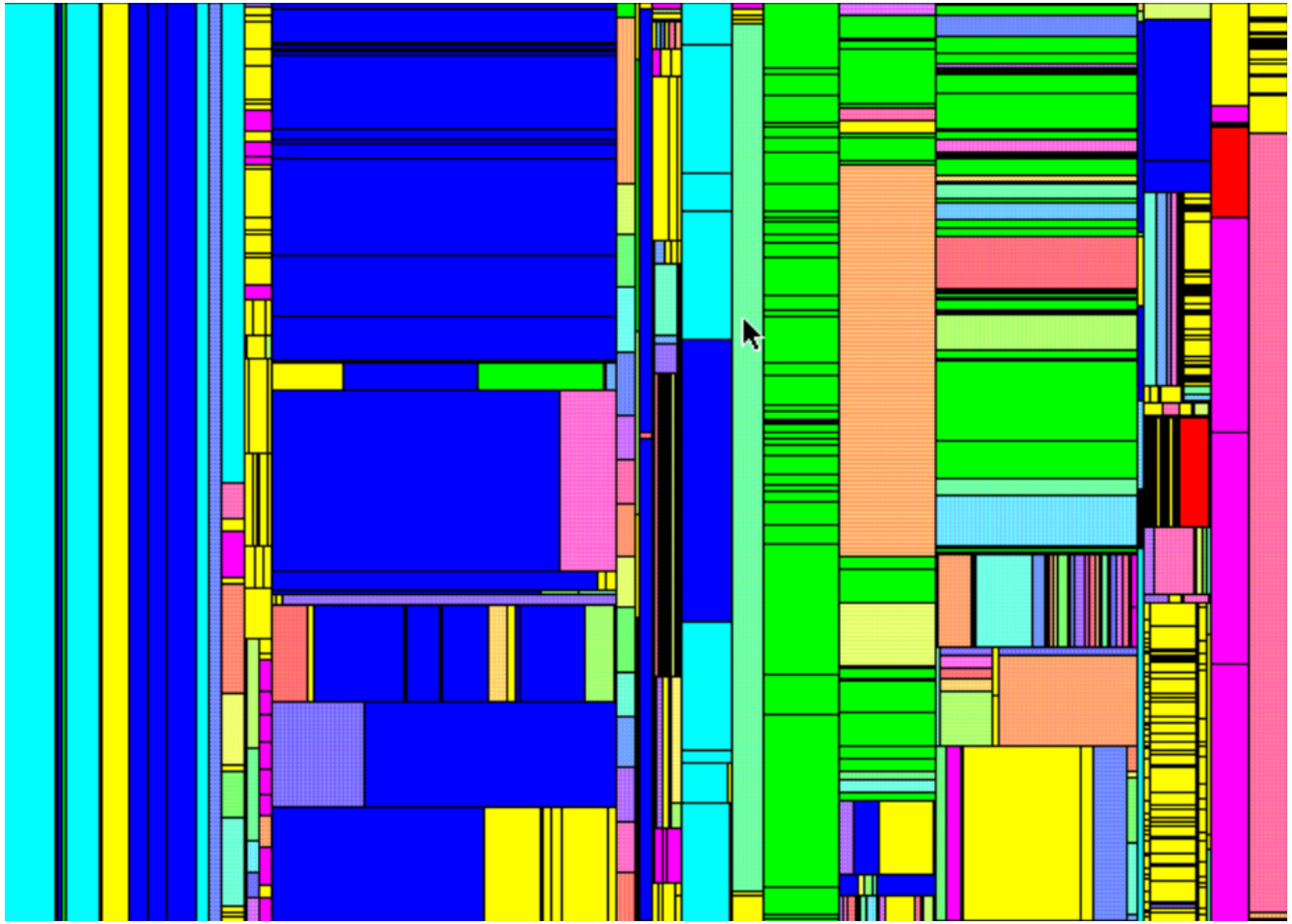
1. Compute sizes (bottom up)
2. Draw boxes (top down based on ratios)

Let s_i be the size of the i-th child of a node.

Then the total size s is $\boldsymbol{s} = \sum_{\boldsymbol{i}=1}^{\boldsymbol{n}} \boldsymbol{s_i}$

and the ratio of the box for the i-th child is $\boldsymbol{r_i} = \frac{\boldsymbol{s_i}}{\boldsymbol{s}}$.

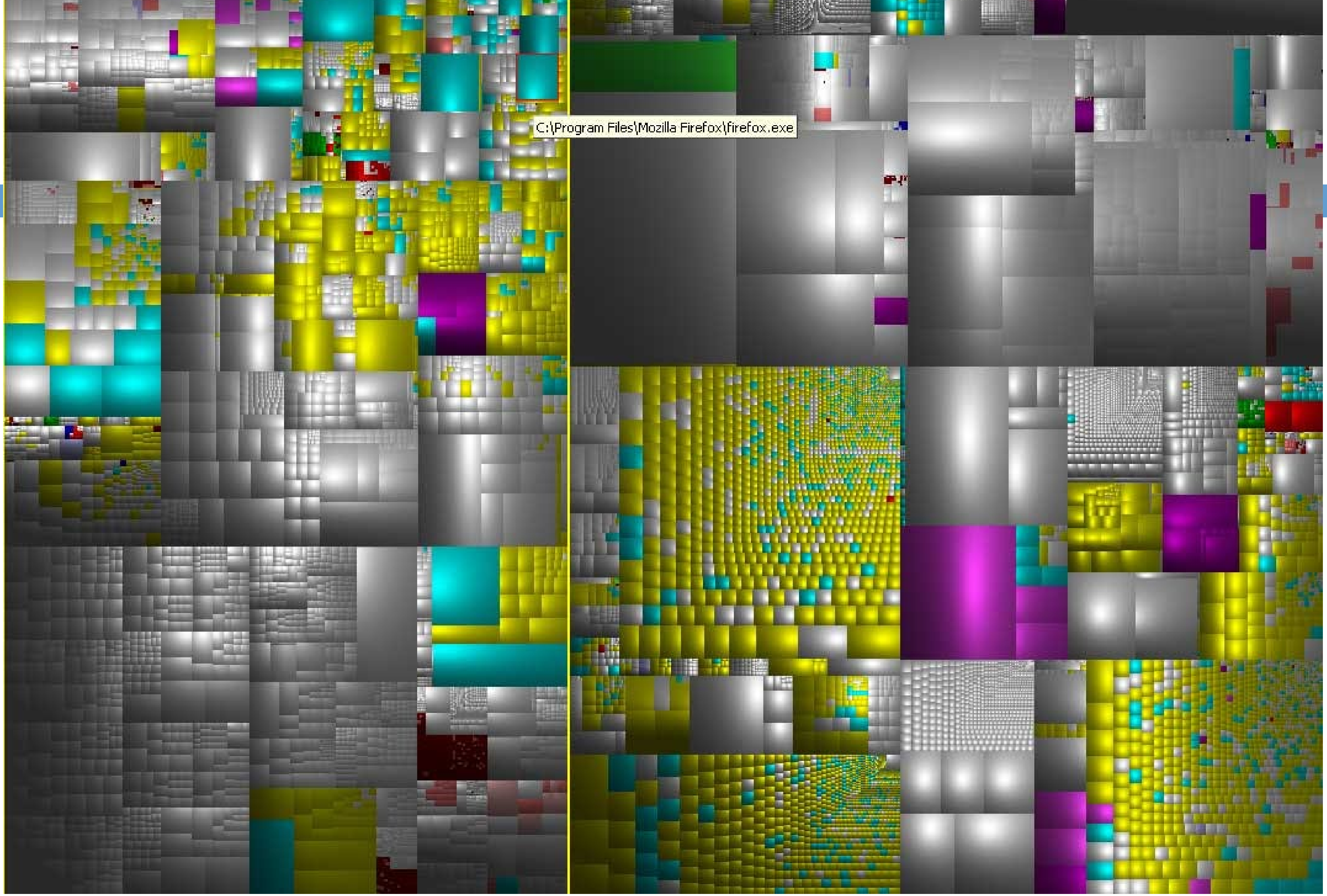
The width and height of the boxes are computed as shown above.



Seminal Paper:
Shneiderman, B. (1992). **Tree visualization with tree-maps: 2-d space-filling approach.** *ACM Transactions on Graphics (TOG)*, *11*(1), 92–99.
https://doi.org/10.1145/102377.115768

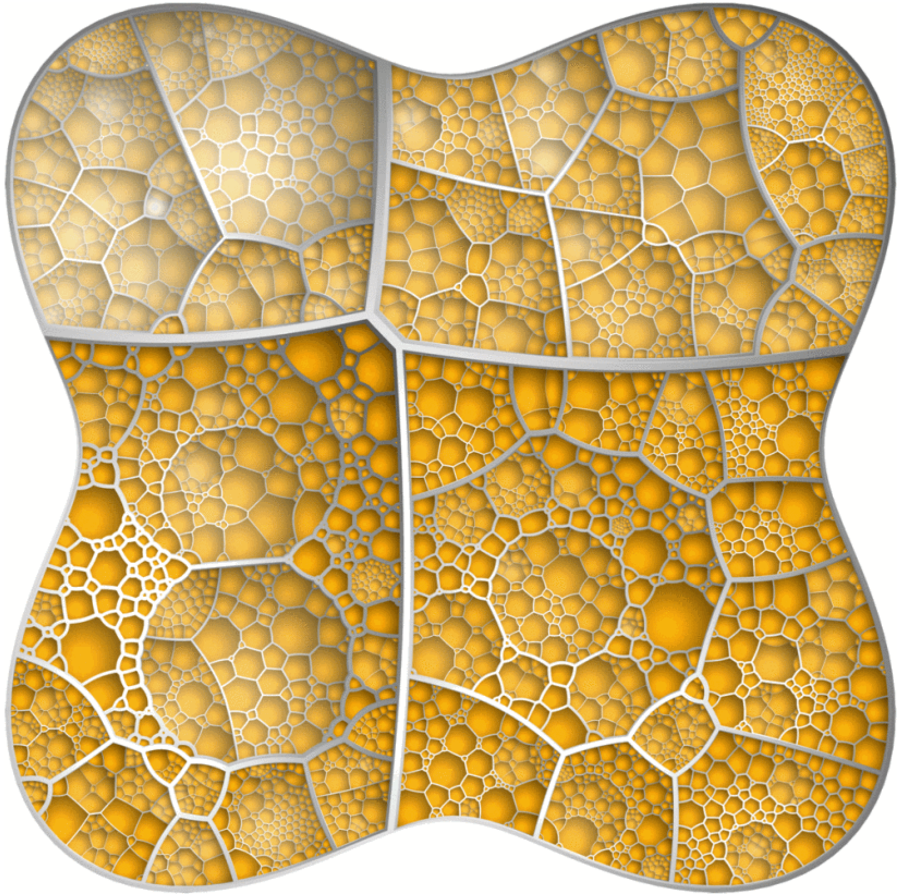
Shneiderman's original implementation was called **TreeViz**, a tool designed specifically to hunt down large files on his laboratory's shared hard drives.

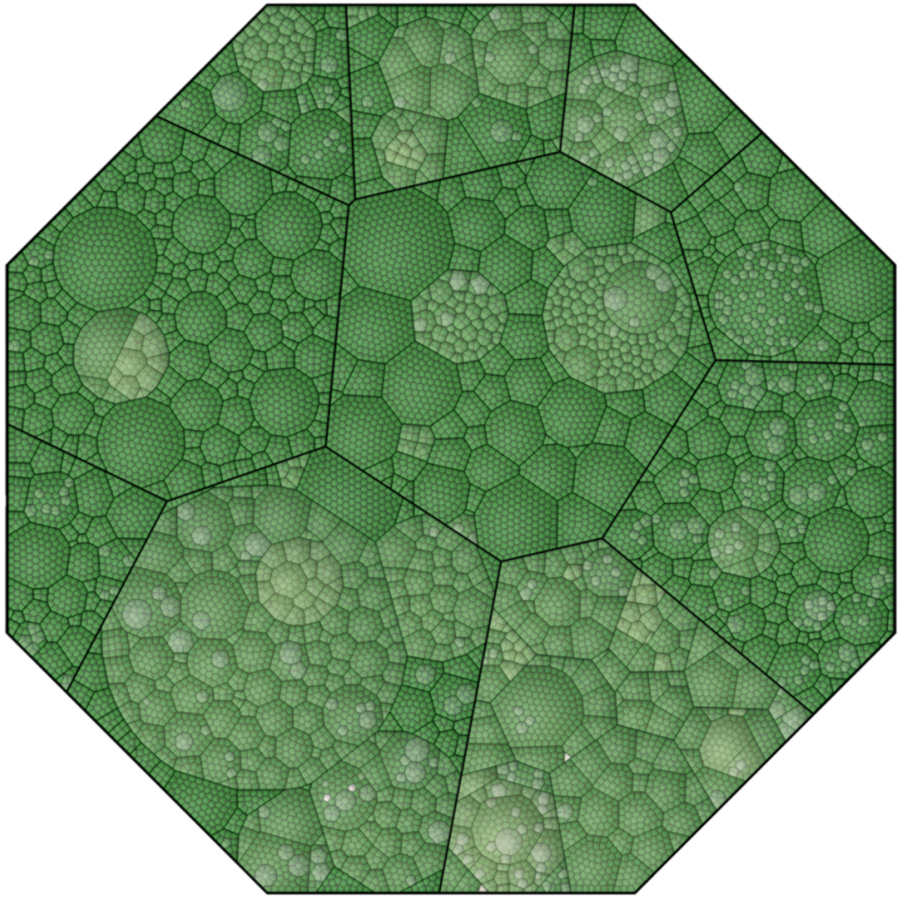
Treemap; Shneiderman, 1992



Cushion Treemap; van Wijk and Wetering, 1999 (Screenshot: SequoiaView)

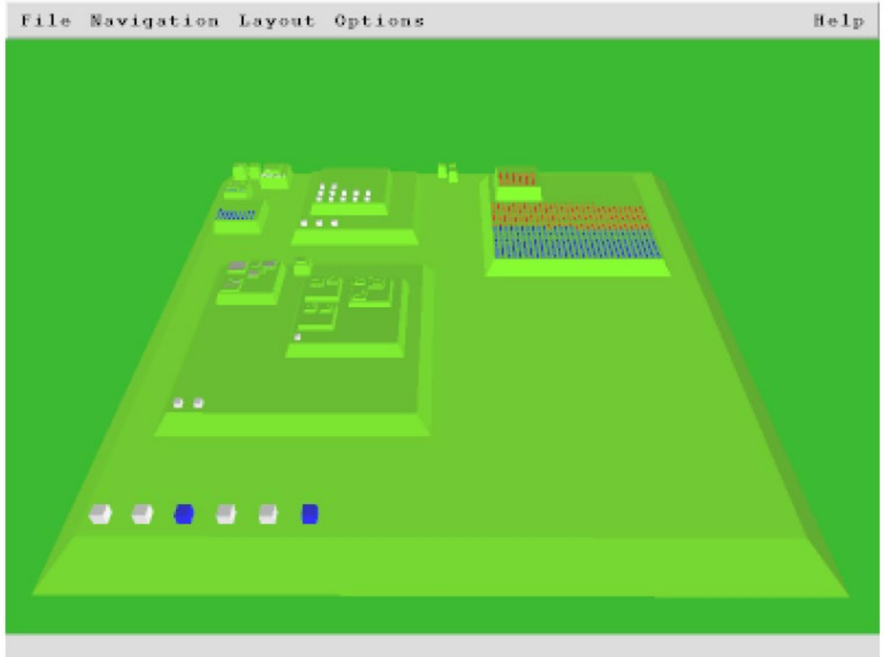
Fill a given shape



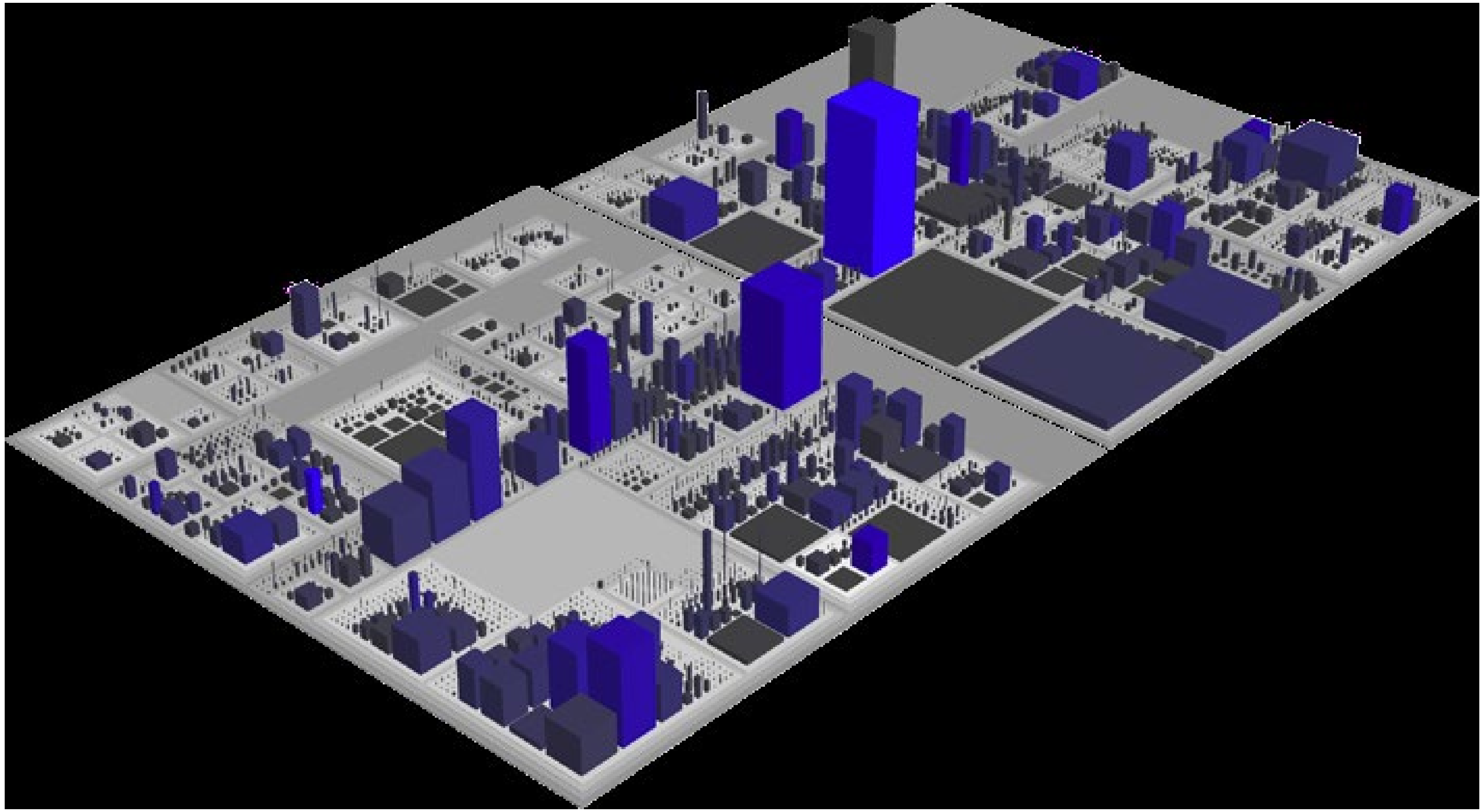


Voronoi treemaps
Balzer and Deussen, INFOVIS 2005

Use 3rd dimension



Information Pyramids
Andrews et al., 1997



CodeCity;
Wettel and Lanza, 2007
(visualization of the JDK)

https://metricity.si.usi.ch/#/evolution/square/tape/all

Discussion: Treemap

Advantages	Disadvantages
Area of leaf nodes can be used	Less intuitive (more intuitive in 3D)
Can fill arbitrary shapes (e.g. Voronoi treemaps)	Hierarchical structure difficult to recognise
Inner nodes require less space	Difficult to add labels
Edges require (almost) no space	

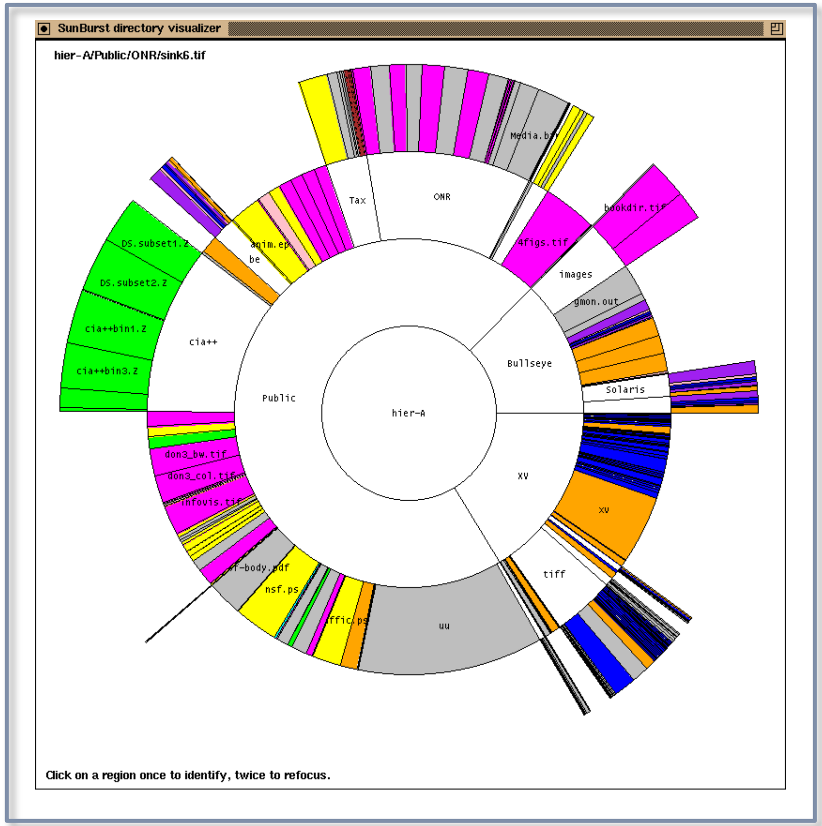
Visual Encoding of Parent-Child Relation

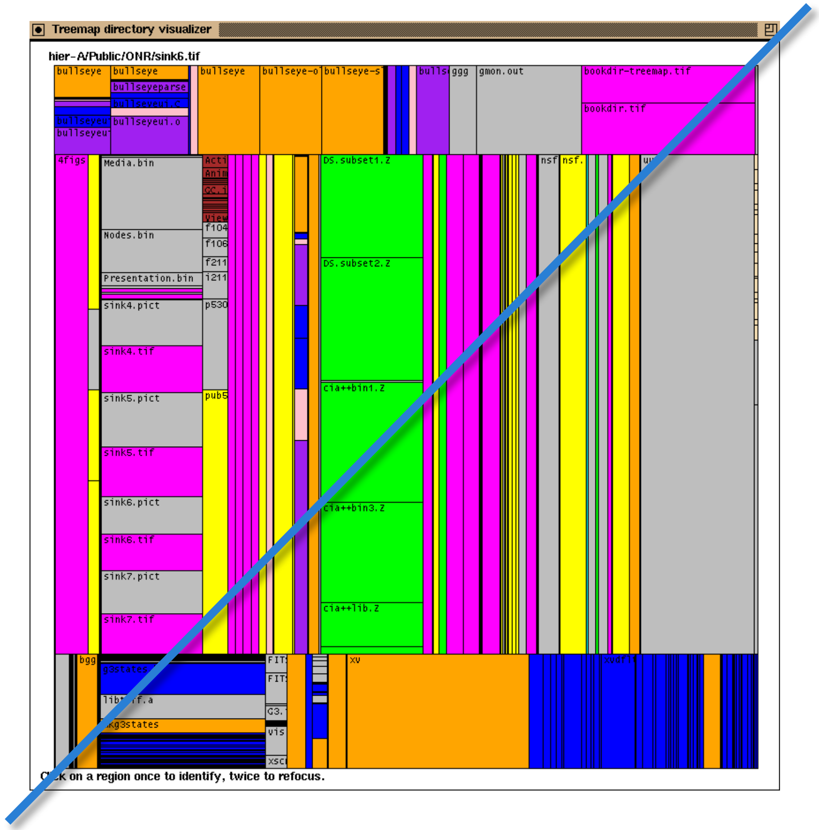
- Node-Link:
  - Connectedness
- Indented Outline Plots:
  - Alignment +(Connectedness)
- Icicle Plots:
  - Proximity
- Treemap:
  - Containment

Chapter 4.6

Some Empirical Results

Sunburst, Treemap





John T. Stasko, Richard Catrambone, Mark Guzdial, Kevin Mcdonald: ***An evaluation of space-filling information visualizations for depicting hierarchical structures.*** Int. J. Hum.-Comput. Stud. 53(5): 663-694 (2000)

Node-Link, Sunburst, Icicle, Treemap

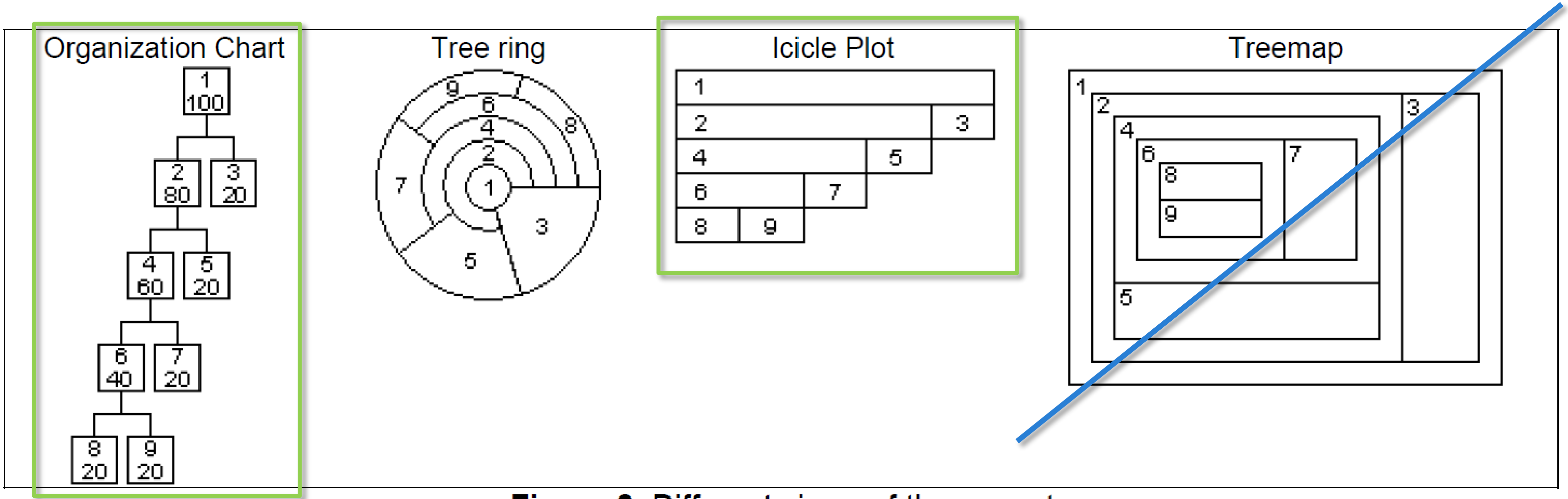
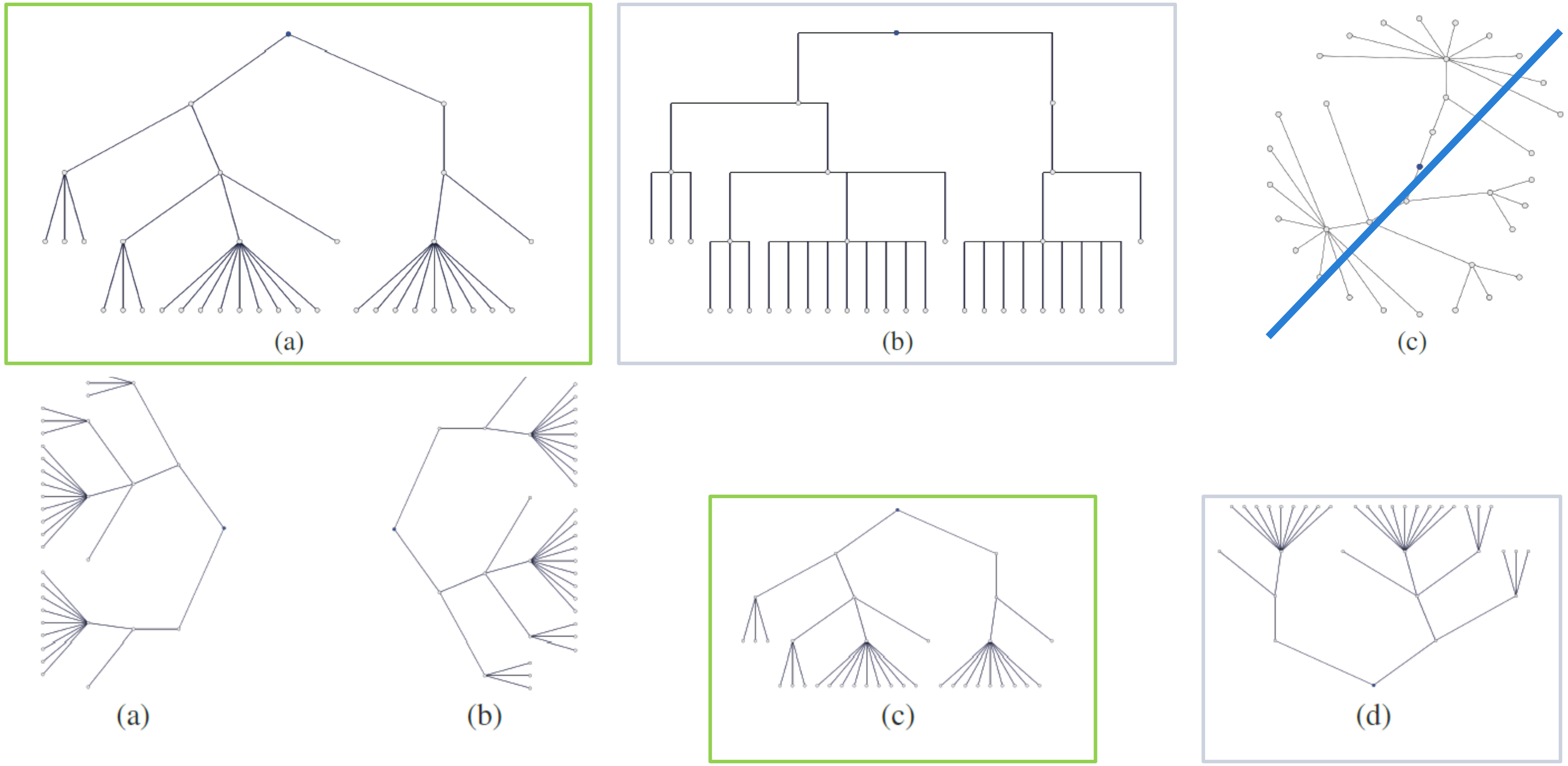


Figure 2. Different views of the same tree

S. Todd Barlow, Padraic Neville: A Comparison of 2-D Visualizations of Hierarchies. INFOVIS 2001: 131-138

Layout of Node-Link Diagrams



Michael Burch, Natalia Konevtsova, Julian Heinrich, Markus Höferlin, Daniel Weiskopf: ***Evaluation of Traditional, Orthogonal, and Radial Tree Diagrams by an Eye Tracking Study.*** IEEE Trans. Vis. Comput. Graph. 17(12): 2440-2448 (2011)

Summary of empirical results

- Recommended
  - Node-Link Diagrams
  - Icicle Plots
  - (Indented Outline)
- Questionable
  - Treemap
  - Radial layouts

- Conclusion
  - Empirical evaluation is just beginning
  - More research is needed to make well-founded design recommendations
  - There is also a lack of domain-specific results.

Chapter 4.6

Visual Comparison of Hierarchies

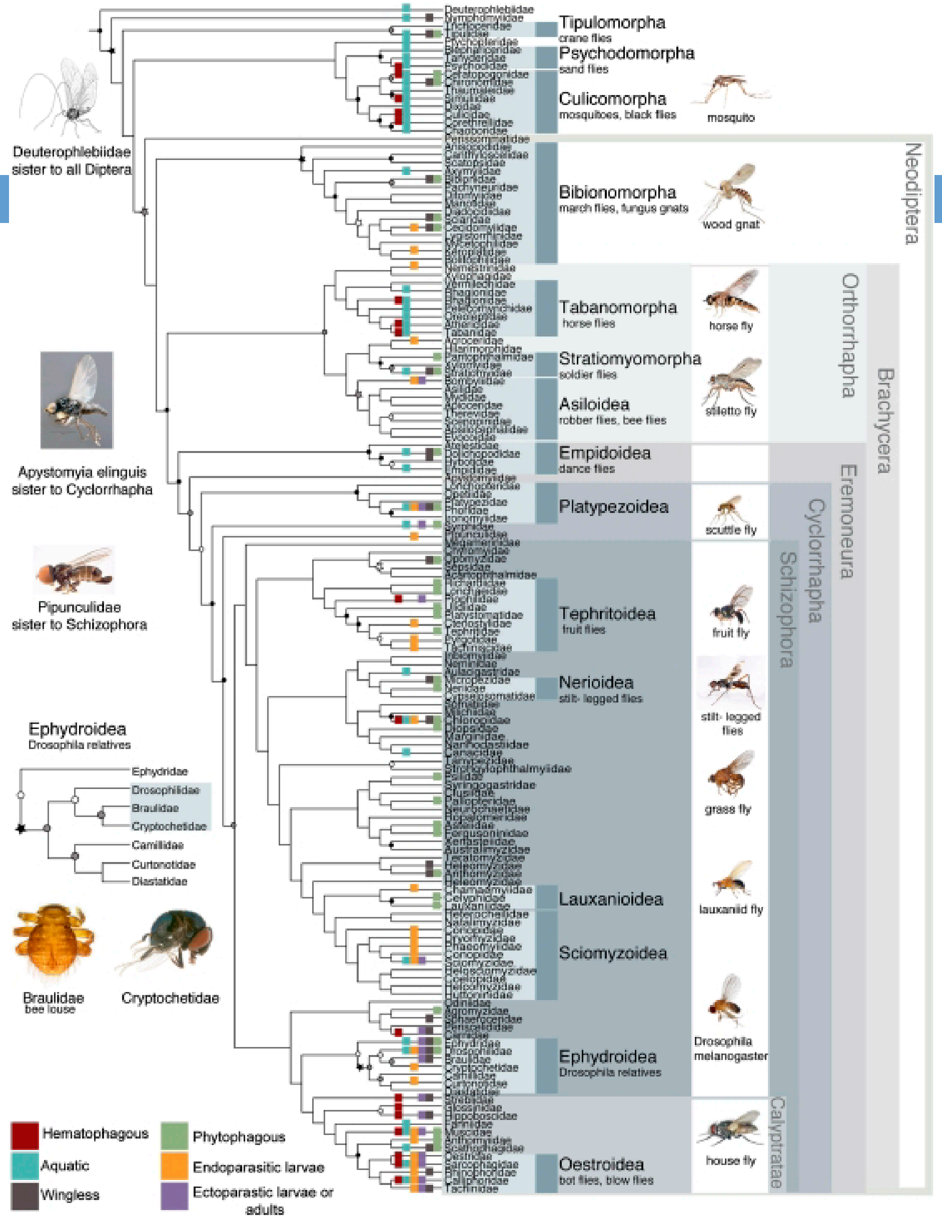
Research Question

Given two hierarchies:

What are the differences and similarities?

Application scenarios:

- Synchronization of file systems
- Temporal change of hierarchies
  - Version differences in software projects
  - Changes in company hierarchy
- Comparison of phylogenetic trees



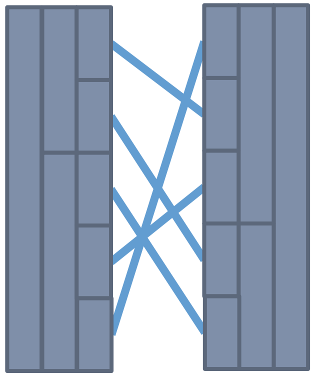
Group work

How would you visually compare two hierarchies? Make a sketch!

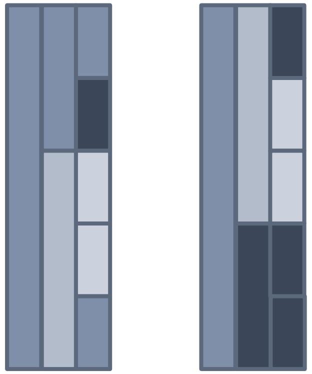
- using …
  - Node-Link Diagrams
  - Indented Outline Plots
  - Icicle Plots
  - Treemaps

Techniques for comparing hierarchies

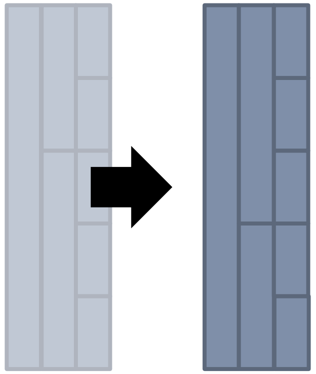
Connecting lines



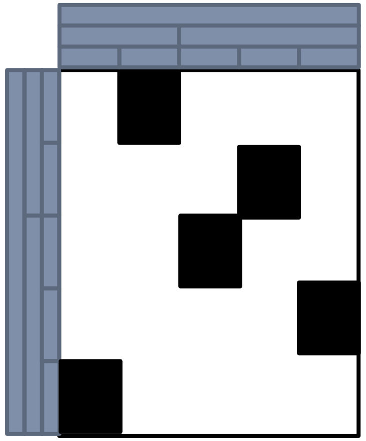
Color



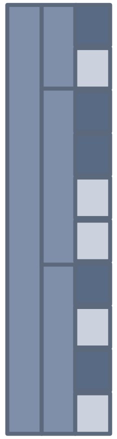
Animation



Matrix



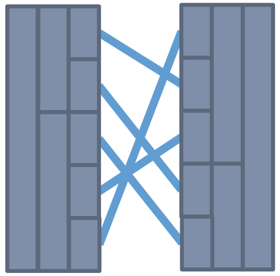
Fusion

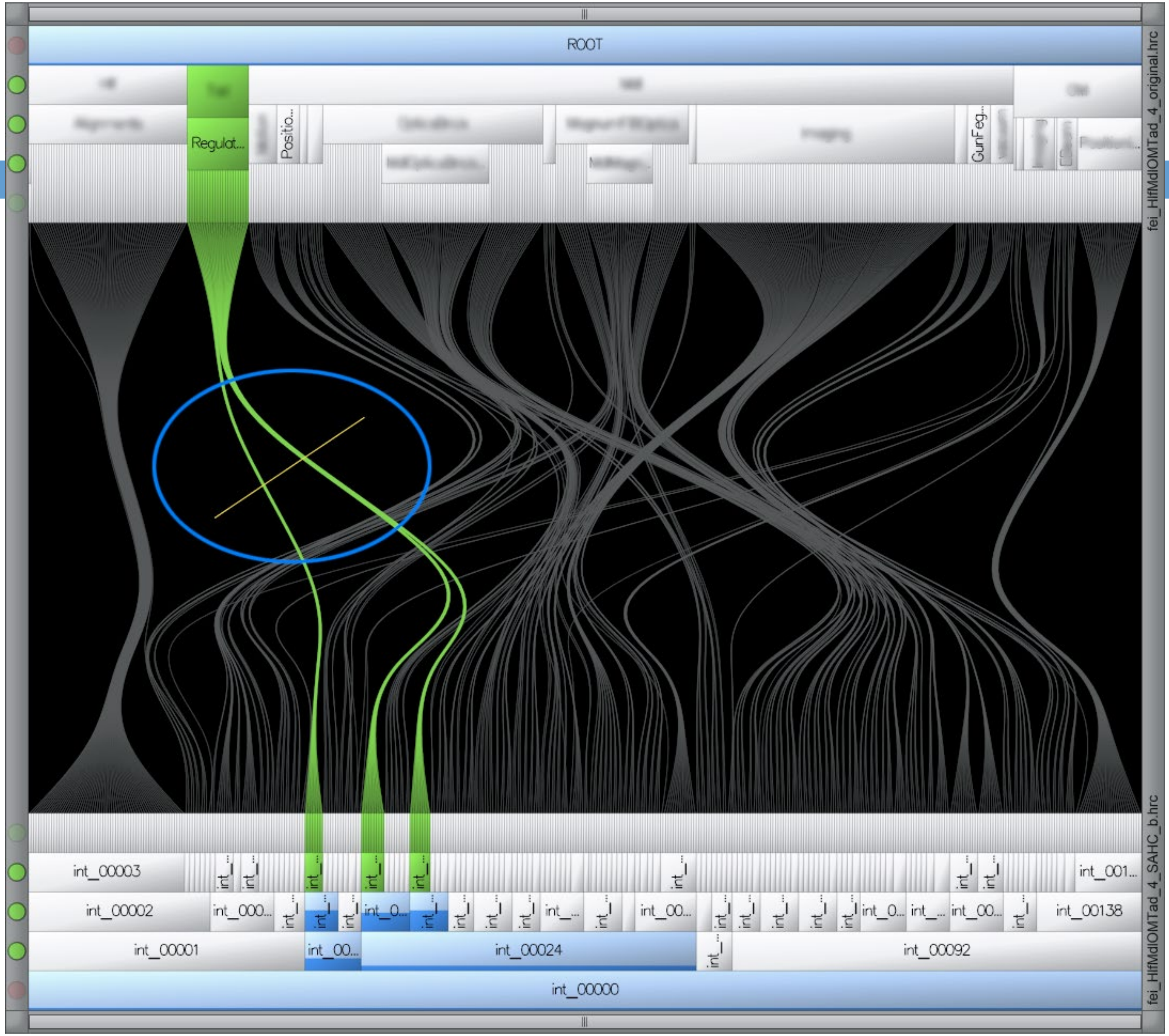


Overlay

???

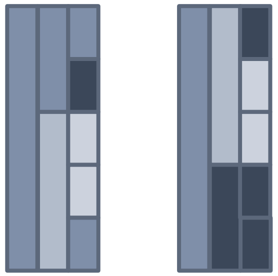
Connecting Lines



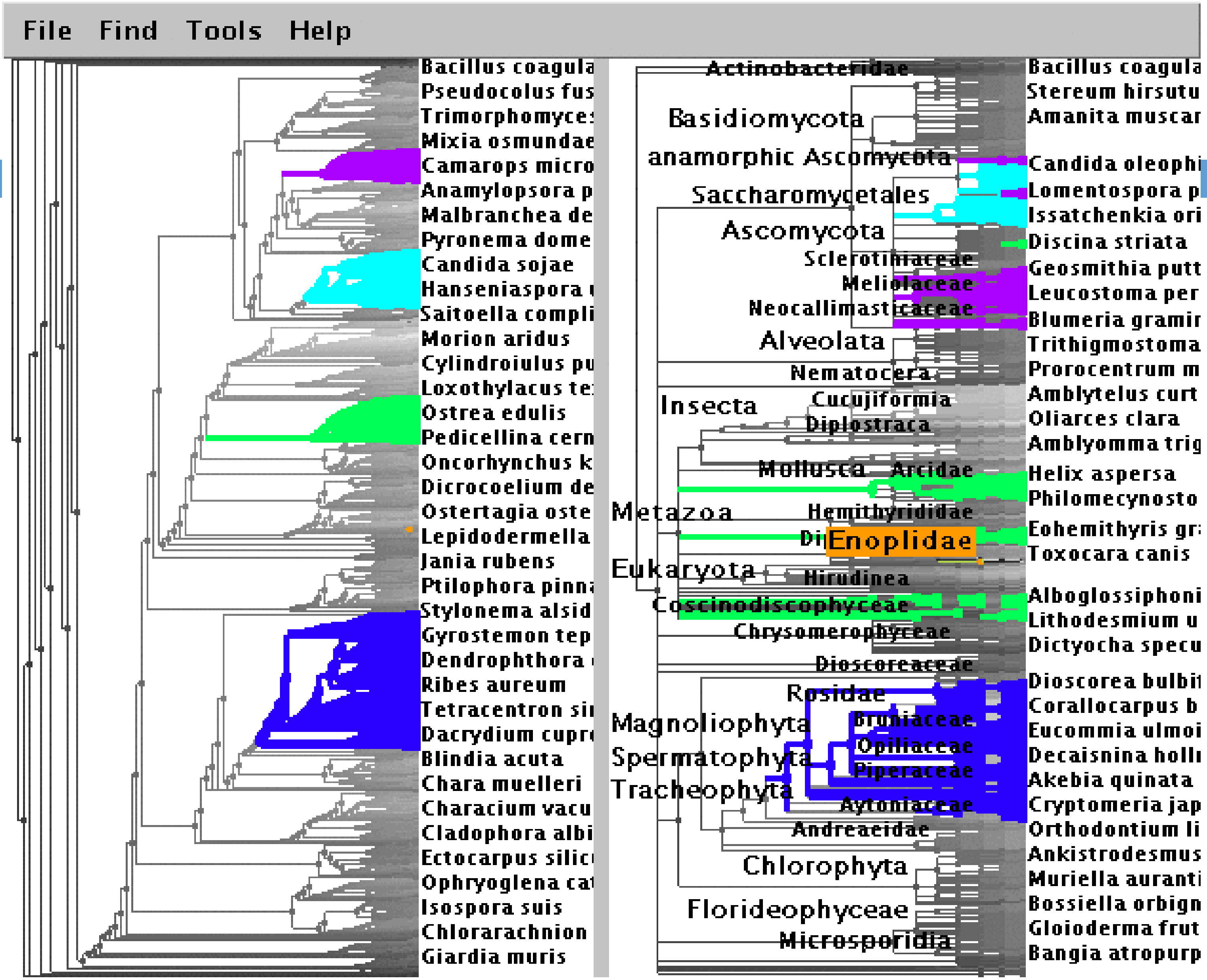


Danny Holten, Jarke J. van Wijk: ***Visual Comparison of Hierarchically Organized Data.*** Comput. Graph. Forum 27(3): 759-766 (2008)

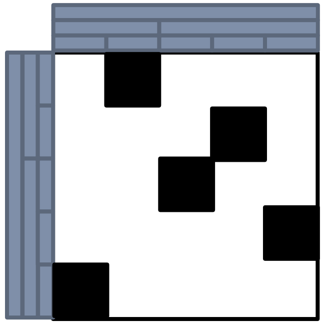
Color

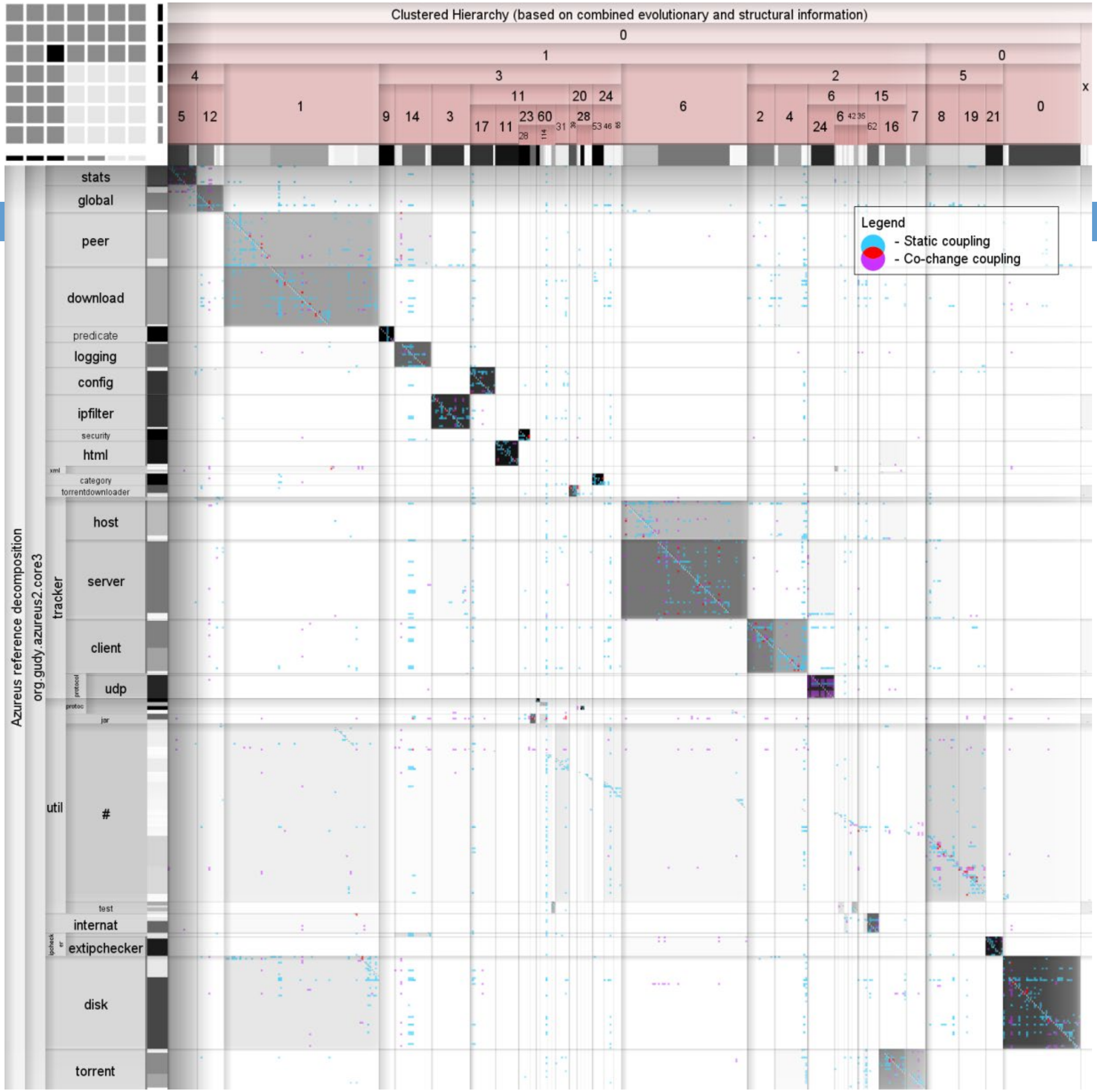


Tamara Munzner, François Guimbretière, Serdar Tasiran, Li Zhang, Yunhong Zhou: ***TreeJuxtaposer: scalable tree comparison using Focus+Context with guaranteed visibility.*** ACM Trans. Graph. 22(3): 453-462 (2003)



(Similarity) Matrix





Fabian Beck, Stephan Diehl: ***Visual comparison of software architectures.*** Information Visualization, April 2013 vol. 12, no. 2, 178-199